Physics 4B Lab 8: *The RC Circuit*

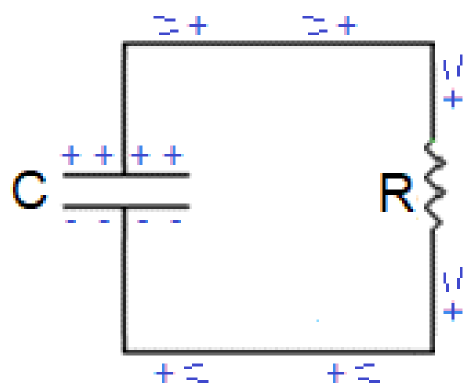
The objective of this lab exercise is to measure and analyze the behavior of an RC circuit.

If a capacitor has some charge on it, the potential across the capacitor is given by the expression:

$$V = \frac{q}{C}$$

where q is the charge on the capacitor and C is the capacitance. Notice that the charge and potential are directly proportional, so they increase or decrease together.

If we connect this capacitor to a resistor to form a simple RC circuit, the capacitor will "discharge" as charge can flow from one side of the capacitor to the other, through the resistor:



We can express the potential on the capacitor as a function of time, and note that this also serves, indirectly, to express the charge on the capacitor as a function of time:

$$V = V_0 \, e^{-t/RC}$$

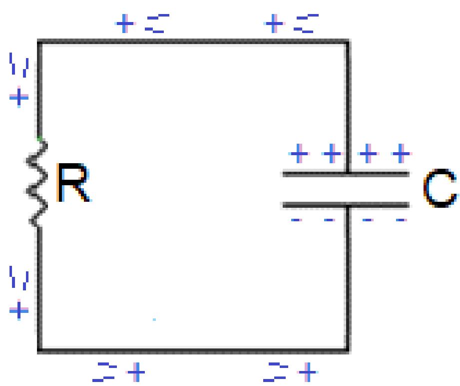
This expression indicates that at time "zero" the potential across the capacitor is some starting value that we call V_0, and as time increases the potential (and charge) decrease exponentially. That is, it decreases rapidly at the beginning, and then the rate of change gradually slows so that at a much later time (when there is very little charge on the capacitor) the rate at which the charge decreases is very slow. (*An appropriate analogy might help: this is like the water level in a tank that has a hole or valve open at the bottom: initially the water flows very quickly, due to the height of the water in the tank, and the level drops quickly; as the water level decreases, the flow of water from the hole also decreases*.)

We can use the equation above as the basis to measure and analyze an RC circuit. Since V_0, R and C are constants for a given circuit, only the potential across the capacitor, V, and time are changing. If we can measure the potential across the capacitor and the corresponding time, we can use the data to measure or compare our constants.

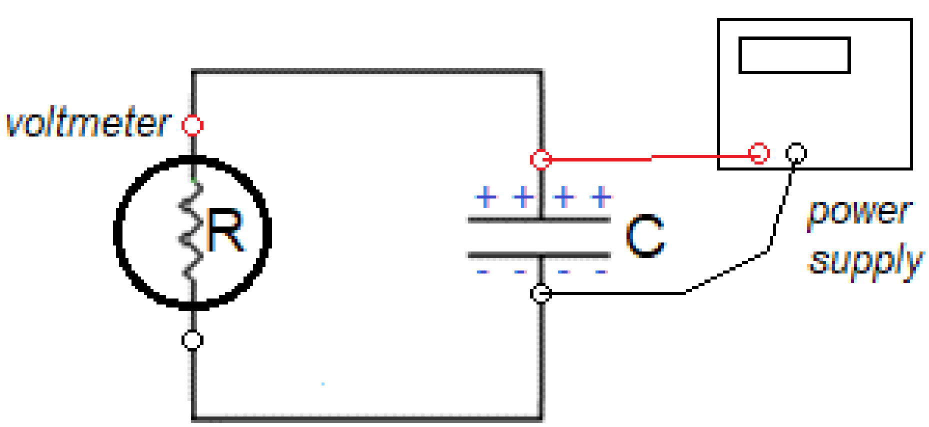
To accomplish this, we will modify the equation slightly by taking the natural log of both sides:

$$lnV = \left(\frac{-1}{RC}\right)t + lnV_0$$

And we can modify our circuit diagram by just flipping it around:



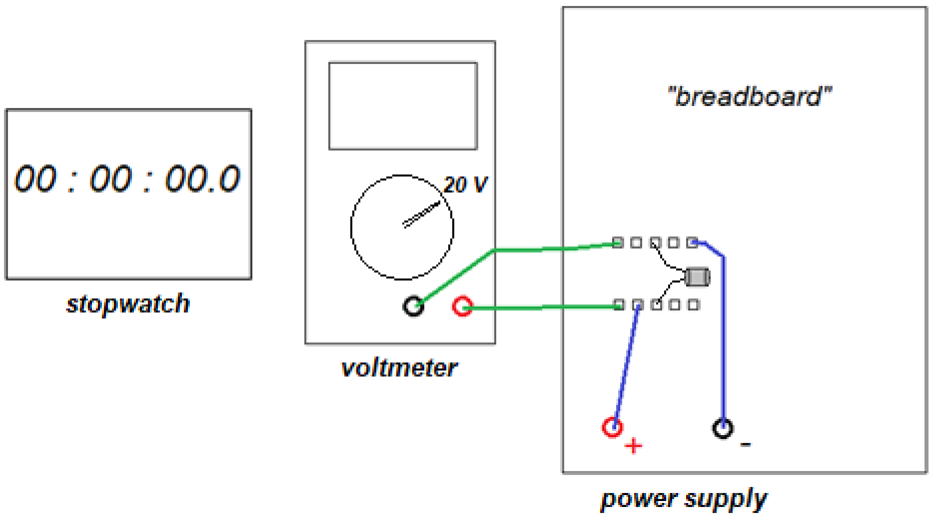
Now we need a way to charge our capacitor and a way to measure the potential across the capacitor. We will charge our capacitor using a power supply, which we will connect (to charge) and then disconnect, so that the capacitor can discharge through the resistor. To measure the potential, we will connect a voltmeter directly to the capacitor. But here we do something very clever: we need a very large resistor so that the capacitor will discharge slowly enough for us to measure. *And voltmeters are designed to have a very large resistance, typically around 10 million Ohms (i.e. around 10 MΩ.)*



When the power supply is disconnected, the capacitor will discharge through the voltmeter (i.e. the "resistor") and the voltmeter will show the changing potential across the capacitor. We can measure and record several values for V and the corresponding time.

The equation above shows that if we create a graph of $\ln V$ vs t, the slope of the best-fit line for this graph should be equal to -1/RC. So if either R or C is known, we can use the slope of the graph to determine the other.

A diagram of our apparatus:



Procedure, Part 1

There are three trials for Part 1, each with a different capacitor. For each trial:

- Connect the voltmeter, capacitor and power supply as shown in the diagram.
- Reset the stopwatch to zero; set the power supply to 12 Volts (as shown on the voltmeter)
- Disconnect the power supply wires from the capacitor and start the stopwatch.
- The capacitor will discharge quickly, as shown by the voltmeter display. **Record the process on video until the reading on the voltmeter drops below 5 Volts.**
- **Review the video and pause when** the voltmeter reads approximately 11.0, 10.5, 10.0, 9.5,…, 6.0, 5.5, 5.0 Volts. At each of these, record the reading on the voltmeter and the time, in seconds.

Create a data table for each of the three trials with columns for the potential of the capacitor, the time in seconds, and the natural log of the potential.

Create a graph of *lnV vs t* for each trial. **All three plots should be shown on one graph**, just as we did for the *Ohm's Law* data for the three resistors in Lab 5. Add an appropriate legend, a best-fit line to each set of data, and the equation for each of the three best-fit lines.

Calculations, Part 1

For each of the three trials, use the value of R, the resistance of the voltmeter, and the measured slope to calculate the value of the capacitance for each of the three capacitors. To accomplish this, we know that, from the equation above:

$$slope = \frac{-1}{RC} \qquad \text{so:} \qquad C = \frac{-1}{R\,(slope)}$$

Note that the product of the SI units of capacitance and resistance must give the SI unit of time:

$$(Farad)(Ohm) \rightarrow seconds \qquad \text{so} \qquad \frac{second}{Ohm} \rightarrow Farad$$

Since we are measuring capacitance in μF, we can use μF in our calculation and the result will be:

$$\frac{sec}{M\Omega} \rightarrow \frac{sec}{\times 10^{6}\Omega} \rightarrow \times 10^{-6}\,F \rightarrow \mu F$$

That is, using *megaOhms* in your calculation will result in *microFarads* for the unit of capacitance.

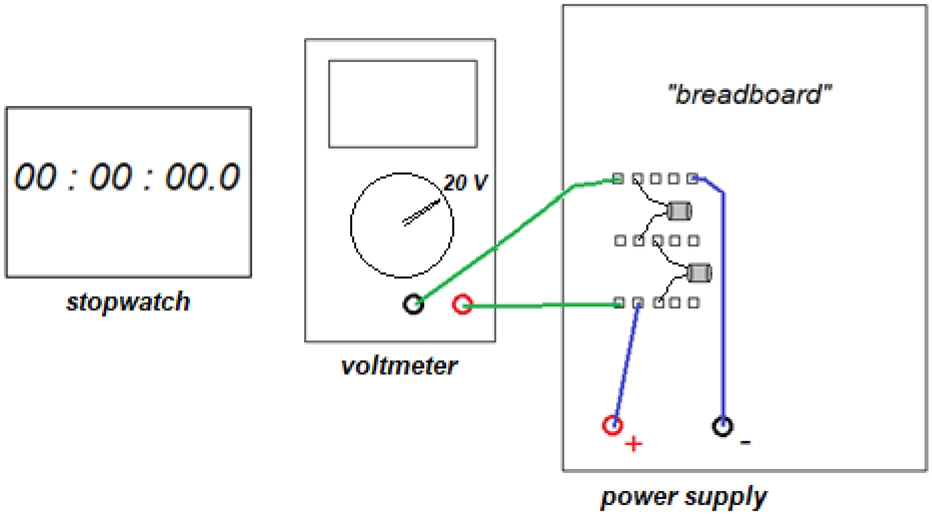
Procedure, Part 2

For Part 2, we will test the theory that the total capacitance of two capacitors in series should be given by the expression:

$$C_{total} = \frac{C_1\,C_2}{C_1 + C_2}$$

The procedure for Part 2 is identical to that of Part 1, except that we will use two capacitors connected in series.

- ***Use the two capacitors with the smallest values of C for Part 2***
- ***Connect the capacitors as shown in the diagram below***



Measure the data for Part 2 exactly the same way as you did for Part 1. Create one data table, and one corresponding graph, with just the one set of data, for Part 2.

Calculations, Part 2

In the ***Calculations*** section, first calculate the expected value of the capacitance, using the two individual values for your capacitors from Part 1. Then use the slope of the graph and the value of R to calculate the *measured* value of the capacitance.

Your expected and measured values of the capacitance should be very close; calculate the percent difference, as usual. (Which means not only have you verified the theory of the RC circuit, but you will have also confirmed the theory of capacitors in series.)